

Gravity: GNSS Positioning Module High Precision GNSS Wiki - DFRobot

SKU:TEL0157 (<https://www.dfrobot.com/product-2651.html>)

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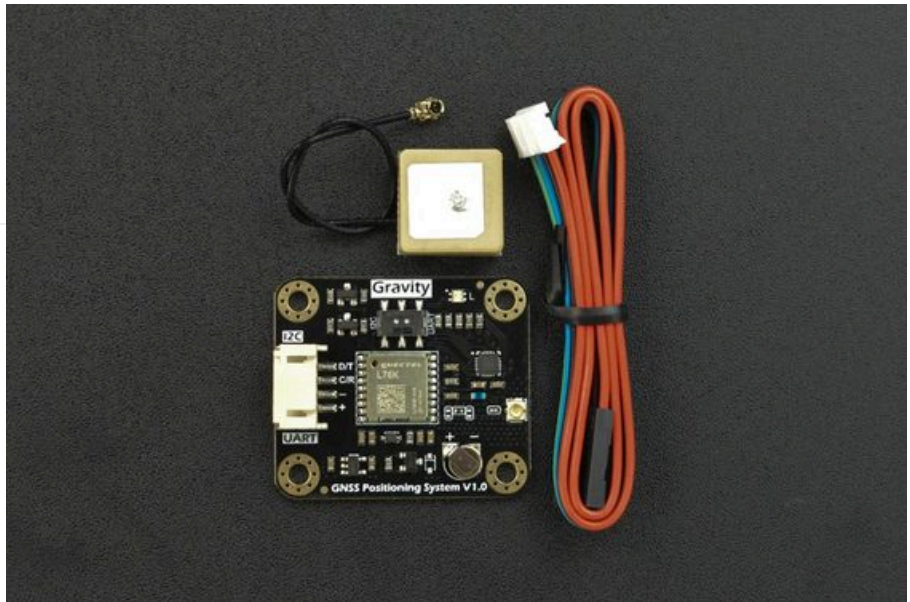
Introduction

Global Navigation Satellite Systems (GNSS) provide critical timing and positioning functions for device operations.

This Gravity: GNSS positioning module from

DFRobot supports both single and multiple systems positioning. It is capable of quick delivery of position data like longitude, latitude, altitude and time. Compared with traditional single GPS positioning, the multi-system combination embraces higher precision and faster speed thanks to the increased number of visible and available satellites, which ensures stable and accurate performance even in complex urban environments.

With I2C and UART data outputs, the GNSS positioning module works well with main-controllers like Arduino, ESP32, and Raspberry Pi. It is applicable to outdoor positioning scenarios such as vehicle navigation, handheld positioning tracker, item tracking and weather station.

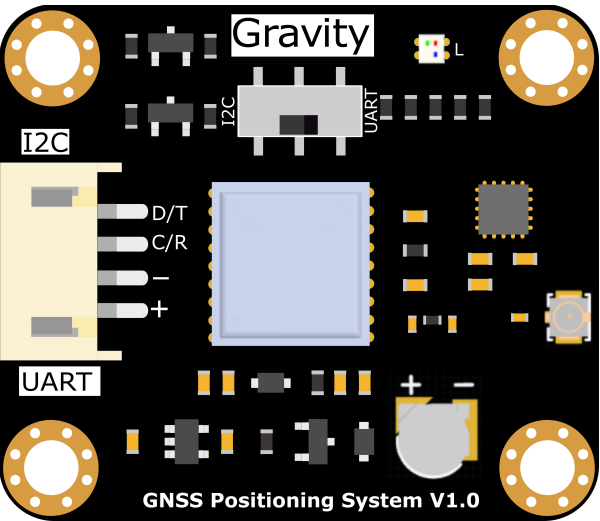


Specification

- Operating Voltage: 3.3 to 5.5V DC
- Output Signal: I2C/UART
- GNSS module : Quectel-L76K
- GNSS bands:

- GPS L1 C/A: 1575.42 MHz
- GLONASS L1: 1602 MHz
- BeiDou B1: 1561.098 MHz
- Channels: 32 tracking channels
- Sensitivity:
 - Auto-aquisition: -147dBm
 - Tracking: -162dBm
 - Re-acquisition: -159dBm
- Accuracy:
 - Position: 2.0m CEP
 - Velocity: 0.1m/s
 - Acceleration: 0.1m/s²
 - Timing: 30ns
- Time for First Positioning: 30s for cold start; 2s for hot start
- Power Consumption: 40mA
- Antenna Interface: IPEX1
- Antenna Frequency: 1561-1575.42MHz±3MHz
- Dimension: 27mm×37mm/1.06×1.46"

Board Overview



Num	Label	Description
1	D/T	I2C data line SDA/UART Data Transmitting- TX
2	C/R	I2C clock line SCL/UART Data Receiving- RX
3	-	GND

4	+	VCC
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When no GPS signal is acquired, the indicator light appears red.

When GPS signal is successfully acquired, the indicator light is green.

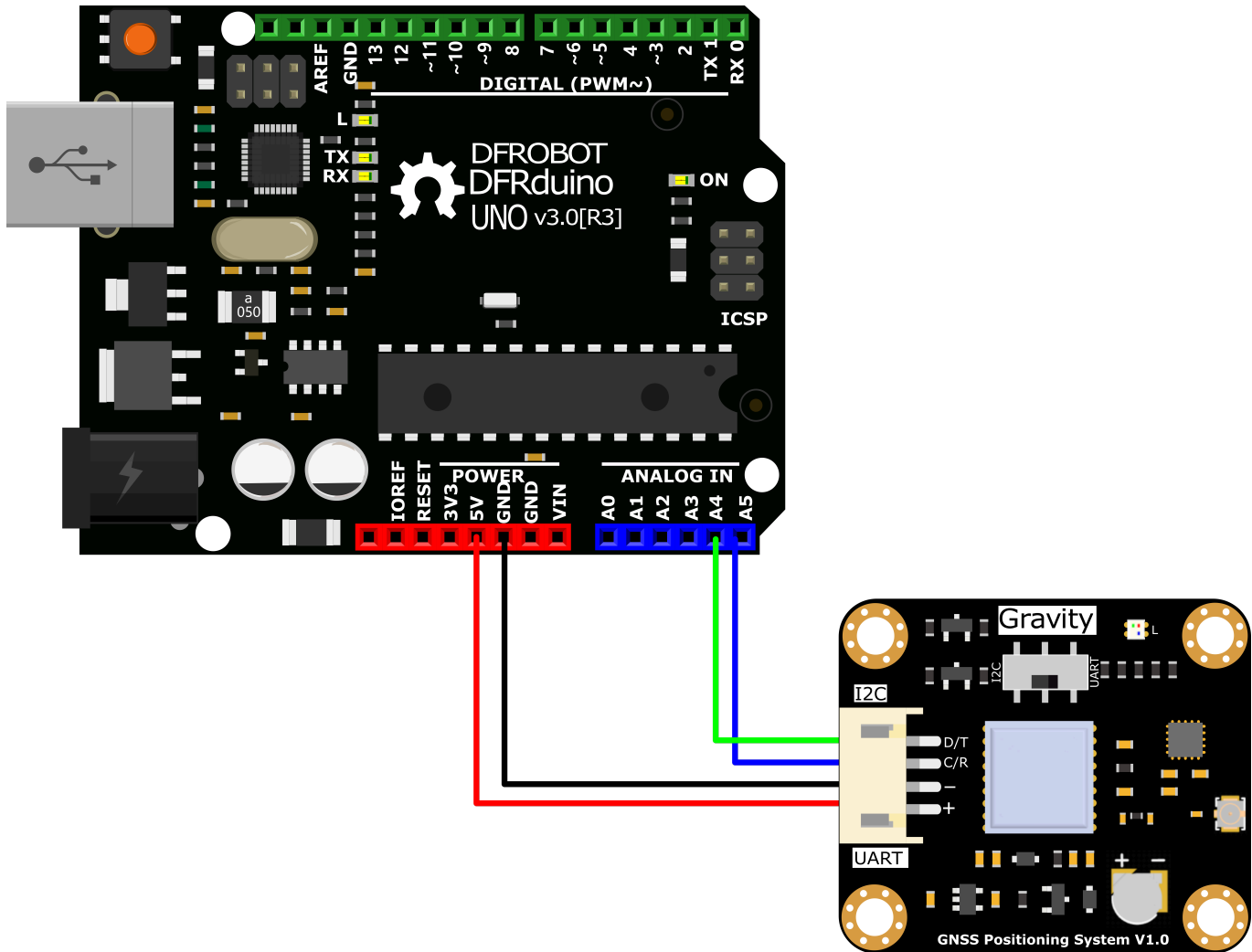
Tutorial for Arduino

Requirements

- **Hardware**
 - DFRduino UNO R3 (<https://www.dfrobot.com/product-838.html>) (or similar) x 1
 - Gravity: GNSS Positioning Module x 1
- **Software**
 - Arduino IDE (<https://www.arduino.cc/en/Main/Software>)
 - Download and install the **DFRobot GNSS Library** (https://github.com/DFRobot/DFRobot_GNSS). (About how to install the library? (<https://www.arduino.cc/en/Guide/Libraries#UxU8mdzF9H0>))

Read Data via I2C

Connection Diagram



Sample Code

- Connect the module to Arduino according to the connection diagram above. It can also be used with Gravity I/O expansion board to prototype ideas faster.
- Change the select switch on the sensor to I2C.
- Download and install the DFRobot_GNSS library (https://github.com/DFRobot/DFRobot_GNSS). (About how to install the library? (<https://www.arduino.cc/en/Guide/Libraries#UxU8mdzF9H0>))
- Open Arduino IDE and upload the following code to Arduino UNO.

- Open the serial monitor of Arduino IDE, set the baud rate to 115200, and observe the printed result.

```
#include "DFRobot_GNSS.h"

DFRobot_GNSS_I2C gnss(&Wire ,GNSS_DEVICE_ADDR);
/*
#ifdef ESP_PLATFORM
    // ESP32 user hardware uart
    // RX io16
    // TX io17
    DFRobot_GNSS_UART gnss(&Serial2 ,9600);
#else
    // Arduino user software uart
    // RX io10
    // TX io11
    SoftwareSerial mySerial(10 ,11);
    DFRobot_GNSS_UART gnss(&mySerial ,9600);
#endif
*/
void setup()
{
    Serial.begin(115200);
    while(!gnss.begin()){
        Serial.println("NO Devices !");
        delay(1000);
    }

    gnss.enablePower();

    /** Set the galaxy to be used
    *   eGPS           USE gps
    *   eBeiDou        USE beidou
    *   eGPS_BeiDou     USE gps + beidou
    *   eGLONASS        USE glonass
    *   eGPS_GLONASS    USE gps + glonass
    *   eBeiDou_GLONASS USE beidou +glonass
    *   eGPS_BeiDou_GLONASS USE gps + beidou + glonass
    */
    gnss.setGnss(eGPS_BeiDou_GLONASS);

    // gnss.setRgbOff();
    gnss.setRgbOn();
    // gnss.disablePower();
}
```

```
void loop()
{
  sTim_t utc = gnss.getUTC();
  sTim_t date = gnss.getDate();

  sLonLat_t lat = gnss.getLat();
  sLonLat_t lon = gnss.getLon();
  double high = gnss.getAlt();
  uint8_t starUserd = gnss.getNumSatUsed();
  double sog = gnss.getSog();
  double cog = gnss.getCog();

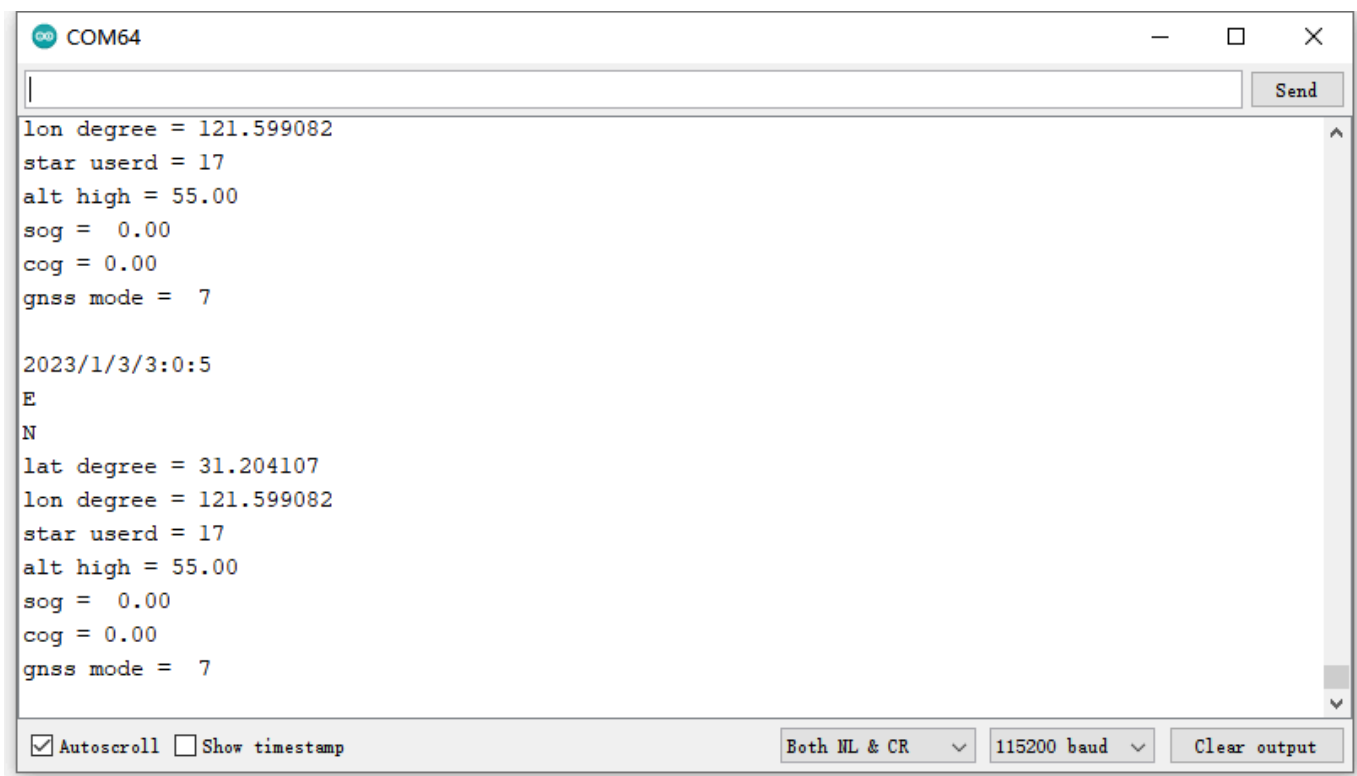
  Serial.println("");
  Serial.print(date.year);
  Serial.print("/");
  Serial.print(date.month);
  Serial.print("/");
  Serial.print(date.date);
  Serial.print("/");
  Serial.print(utc.hour);
  Serial.print(":");
  Serial.print(utc.minute);
  Serial.print(":");
  Serial.print(utc.second);
  Serial.println();
  Serial.println((char)lat.latDirection);
  Serial.println((char)lon.lonDirection);

  // Serial.print("lat DDMM.MMMMM = ");
  // Serial.println(lat.latitude, 5);
  // Serial.print(" lon DDDMM.MMMMM = ");
  // Serial.println(lon.longitude, 5);
  Serial.print("lat degree = ");
  Serial.println(lat.latitudeDegree,6);
  Serial.print("lon degree = ");
  Serial.println(lon.longitudeDegree,6);

  Serial.print("star userd = ");
  Serial.println(starUserd);
  Serial.print("alt high = ");
  Serial.println(high);
  Serial.print("sog = ");
  Serial.println(sog);
  Serial.print("cog = ");
  Serial.println(cog);
  Serial.print("gnss mode = ");
  Serial.println(gnss.getGnssMode());
  delay(1000);
}
```

Result

Open serial monitor to see the result.



The screenshot shows a serial monitor window titled "COM64". The window has a text input field at the top with a "Send" button. The main area displays the following text:

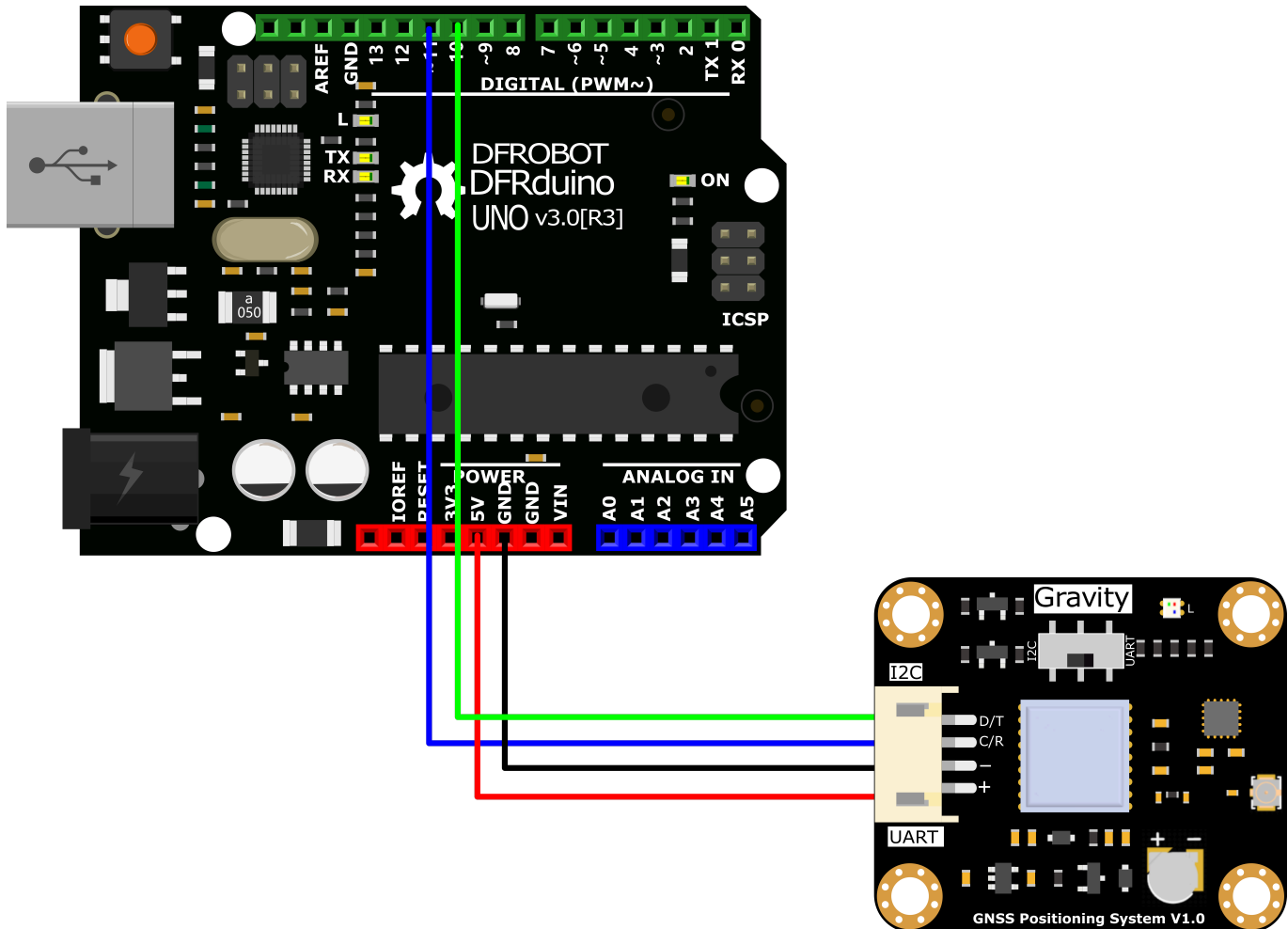
```
lon degree = 121.599082
star userd = 17
alt high = 55.00
sog = 0.00
cog = 0.00
gnss mode = 7

2023/1/3/3:0:5
E
N
lat degree = 31.204107
lon degree = 121.599082
star userd = 17
alt high = 55.00
sog = 0.00
cog = 0.00
gnss mode = 7
```

At the bottom of the window, there are several controls: a checkbox for "Autoscroll" (checked), a checkbox for "Show timestamp" (unchecked), a dropdown menu set to "Both NL & CR", a dropdown menu set to "115200 baud", and a "Clear output" button.

Read Data via UART

Connection Diagram



Sample Code

- Connect the module to Arduino according to the connection diagram above. It can also be used with Gravity I/O expansion board to prototype ideas faster.
- Change the select switch on the sensor to UART.
- Download and install the DFRobot_GNSS library (https://github.com/DFRobot/DFRobot_GNSS). (About how to install the library? (<https://www.arduino.cc/en/Guide/Libraries#.UxU8mdzF9H0>))
- Open Arduino IDE and upload the following code to Arduino UNO.
- Open the serial monitor of Arduino IDE, set the baud rate to 115200, and observe the printed result.

printed result.

```
/*!  
 * @file getGNSS.ino  
 * @copyright Copyright (c) 2010 DFRobot Co.Ltd (http://www.dfrobot.com)  
 * @license The MIT License (MIT)  
 * @author ZhixinLiu(zhixin.liu@dfrobot.com)  
 * @version V0.1  
 * @date 2022-08-15  
 * @url https://github.com/dfrobot/DFRobot_GNSS  
 */  
  
#include "DFRobot_GNSS.h"  
  
//DFRobot_GNSS_I2C gnss(&Wire ,GNSS_DEVICE_ADDR);  
  
#ifdef ESP_PLATFORM  
    // ESP32 user hardware uart  
    // RX io16  
    // TX io17  
    DFRobot_GNSS_UART gnss(&Serial2 ,9600);  
#else  
    // Arduino user software uart  
    // RX io10  
    // TX io11  
    SoftwareSerial mySerial(10 ,11);  
    DFRobot_GNSS_UART gnss(&mySerial ,9600);  
#endif  
  
void setup()  
{  
    Serial.begin(115200);  
    while(!gnss.begin()){  
        Serial.println("NO Devices !");  
        delay(1000);  
    }  
  
    gnss.enablePower();
```

```
/** Set the galaxy to be used
 *   eGPS          USE gps
 *   eBeiDou       USE beidou
 *   eGPS_BeiDou   USE gps + beidou
 *   eGLONASS      USE glonass
 *   eGPS_GLONASS  USE gps + glonass
 *   eBeiDou_GLONASS  USE beidou +glonass
 *   eGPS_BeiDou_GLONASS  USE gps + beidou + glonass
 */
gnss.setGnss(eGPS_BeiDou_GLONASS);

// gnss.setRgbOff();
gnss.setRgbOn();
// gnss.disablePower();
}

void loop()
{
    sTim_t utc = gnss.getUTC();
    sTim_t date = gnss.getDate();
    sLonLat_t lat = gnss.getLat();
    sLonLat_t lon = gnss.getLon();
    double high = gnss.getAlt();
    uint8_t starUserd = gnss.getNumSatUsed();
    double sog = gnss.getSog();
    double cog = gnss.getCog();

    Serial.println("");
    Serial.print(date.year);
    Serial.print("/");
    Serial.print(date.month);
    Serial.print("/");
    Serial.print(date.date);
    Serial.print("/");
    Serial.print(utc.hour);
    Serial.print(":");
    Serial.print(utc.minute);
    Serial.print(":");
    Serial.print(utc.second);
```

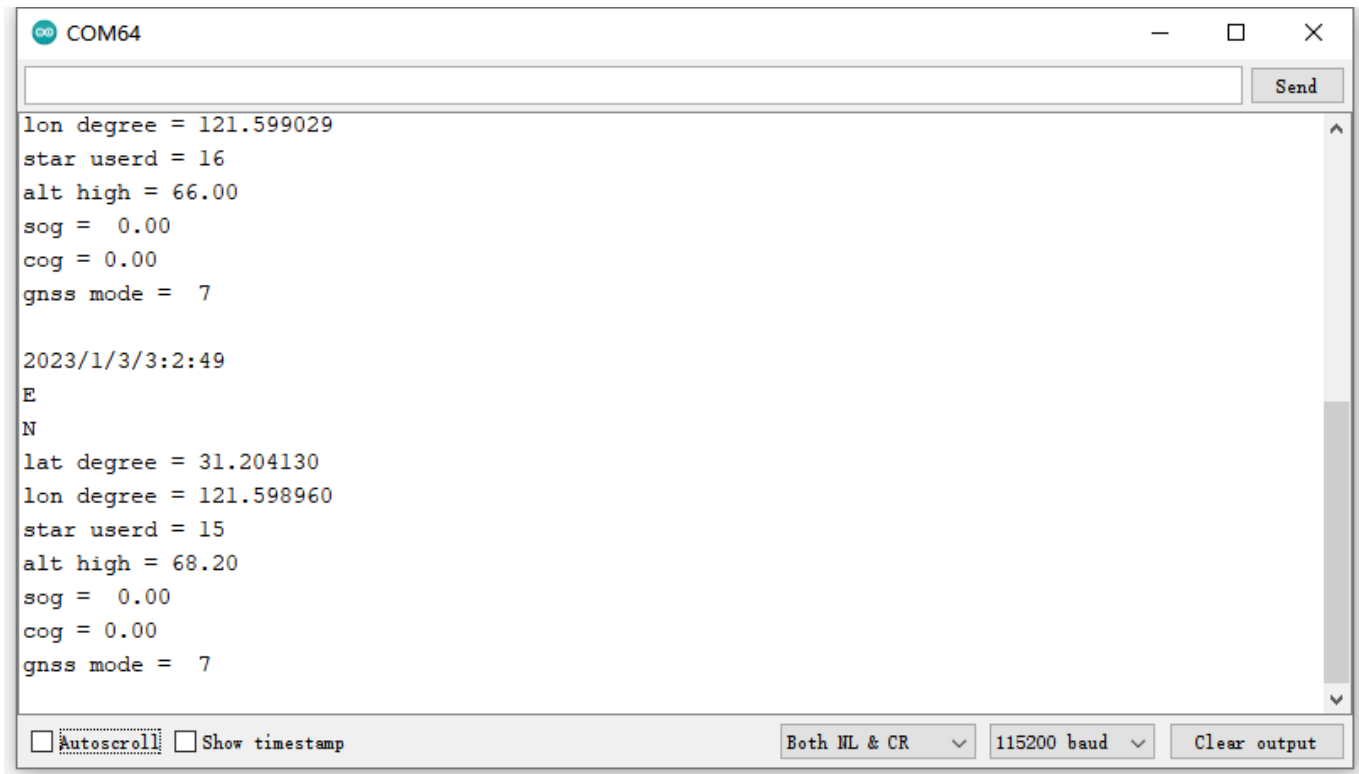
```
Serial.println();
Serial.println((char)lat.latDirection);
Serial.println((char)lon.lonDirection);

// Serial.print("lat DDMM.MMMMM = ");
// Serial.println(lat.latitude, 5);
// Serial.print(" lon DDDMM.MMMMM = ");
// Serial.println(lon.longitude, 5);
Serial.print("lat degree = ");
Serial.println(lat.latitudeDegree,6);
Serial.print("lon degree = ");
Serial.println(lon.longitudeDegree,6);

Serial.print("star userd = ");
Serial.println(starUserd);
Serial.print("alt high = ");
Serial.println(high);
Serial.print("sog = ");
Serial.println(sog);
Serial.print("cog = ");
Serial.println(cog);
Serial.print("gnss mode = ");
Serial.println(gnss.getGnssMode());
delay(1000);
}
```

Result

Open serial monitor to see the result.



FAQ

Q1:

Why can't I get a GPS signal?

A1:

This product only uses GPS functionality and does not use WiFi or Bluetooth for location assistance like mobile phones or other electronic devices. So we recommend that you use this product by a window, on a penthouse, or outdoors.

When the on-board indicator light turns from red to green, it means that the product has successfully acquired a GPS signal.

Q2 : Encountering I2C address conflicts?

A2: For a comprehensive guide on identifying and resolving I2C address conflicts in embedded systems, check out this detailed article: [How to Resolve I2C Address Conflicts](https://wiki.dfrobot.com/How_to_Resolve_I2C_Address_Conflicts_in_Embedded_Systems) (https://wiki.dfrobot.com/How_to_Resolve_I2C_Address_Conflicts_in_Embedded_Systems). It covers practical ha

More questions and cool idea, please visit DFRobot Forum (<https://www.dfrobot.com/forum/>)

More Documents

- Schematics
(<https://dfimg.dfrobot.com/nobody/wiki/347648af182e0024278ae9c33b514125.pdf>)
- Dimensions & Component Layout
(<https://dfimg.dfrobot.com/nobody/wiki/40c2f9f5ba39bd9fee5be7ee3af5b864.pdf>)



Get **Gravity: GNSS GPS BeiDou Receiver Module** (<https://www.dfrobot.com/product-2651.html>) from DFRobot Store or **DFRobot Distributor**. (<https://www.dfrobot.com/distributor>)

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